Where does a self-driving AI fail?

Finding blind spots inside a driving AI's own activations

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BACKGROUND

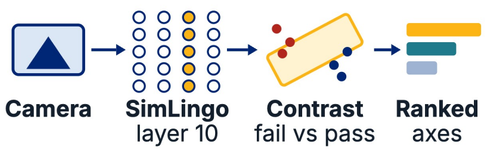
1 Which data should we collect next?

- Driving AI fails on **inputs unlike its training data**
- Test lists only catch **crashes someone imagined**

Test list: night, fog, animals, camera fault
? Failures nobody thought to list

METHODOLOGY

2 Reading the model's own signals



SimLingo drives from one camera. We plant failures on purpose and check what discovery finds.

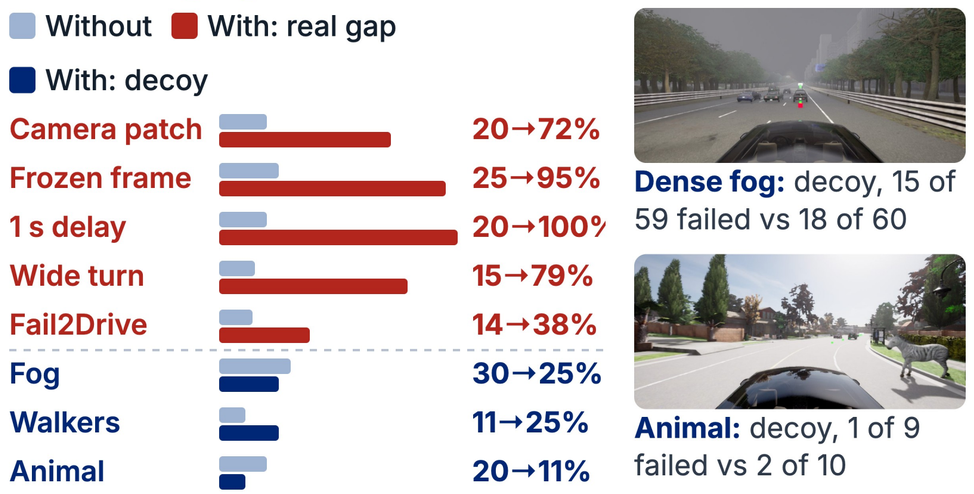
Our runs match the published score



Bench2Drive score, 220 routes (0–100)

RESULTS · PLANTED FAILURES

4 Hiding what the camera sees breaks it; changing looks doesn't



Dense fog: decoy, 15 of 59 failed vs 18 of 60

Animal: decoy, 1 of 9 failed vs 2 of 10

% of paired routes that failed, without → with the change. Dev data.

KEY RESULT · PLANTED FAILURE

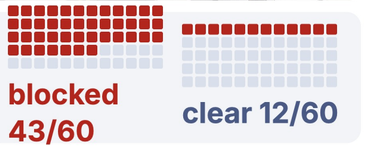
3 Block the camera centre and it hits what it can't see





20% → 72%
routes failed

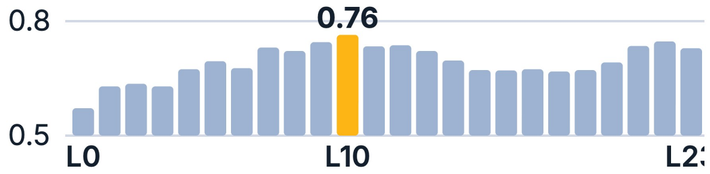
31 : 0
only blocked : only clear



Same 60 routes driven twice. Each square is a route; red = failed (Fisher p = 1.9e-8).

RESULTS · INSIDE THE MODEL

5 Layer 10 flags failures best



AUROC by layer: 212 routes, 61 failing

RESULTS · DISCOVERY METHODS

6 Only camera patches are found

Overlap with true failures (0–1, higher is better). Dev data.

CONCLUSIONS

7 What it means

- Activations find **one failure type**: a visible camera patch
- Fail2Drive: 28 unseen routes fail, 4 the other way; **baseline B3 wins**
- **Next:** blind test pool

REFERENCES
Renz et al., SimLingo, CVPR 2025 · Jia et al., Bench2Drive, NeurIPS 2024 · Dosovitskiy et al., CARLA, CoRL 2017 · Majumdar et al., Predictive Red Teaming, CoRL 2025 · Gerstenecker et al., Fail2Drive, IROS 2026 · Swann et al., Dr. VLA, arXiv 2603.19183

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